

# Ubiquitous Language — CyClaw

## Query Lifecycle

| Term                       | Definition                                                                                                                            | Aliases to avoid                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Query</b>               | A natural-language question submitted by a user to the CyClaw system                                                                  | Prompt, request, user message                              |
| <b>Score Gate</b>          | The <code>min_score</code> threshold that determines whether retrieved context is trustworthy enough to answer locally                | Confidence gate, score threshold (use only as description) |
| <b>Local Answer</b>        | An answer generated by the local LLM using retrieved context from the knowledge base                                                  | Offline answer, on-premise response                        |
| <b>Online Escalation</b>   | Routing a query to an external LLM (Grok or Claude) when local retrieval scores are too low                                           | Fallback, external call, cloud fallback (ambiguous)        |
| <b>User Confirmation</b>   | Explicit human approval required before any query escalates to an online provider                                                     | Confirm, user consent, user gate                           |
| <b>Offline Best-Effort</b> | An answer generated by the local LLM with reduced confidence when the user declines online escalation                                 | Low-confidence answer, degraded answer                     |
| <b>Retrieval Mode</b>      | Which search path produced results: <code>semantic</code> , <code>keyword</code> , <code>hybrid</code> , or degraded variants thereof | Search type, query mode                                    |

## Retrieval & Search

| Term                   | Definition                                                                                                   | Aliases to avoid                         |
|------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Corpus</b>          | The collection of source documents ( <code>.md</code> , <code>.txt</code> ) that comprise the knowledge base | Documents, files, source data            |
| <b>Chunk</b>           | A fixed-size text segment produced by splitting a corpus document; the unit of indexing and retrieval        | Fragment, segment, piece                 |
| <b>Semantic Search</b> | Vector-similarity search over chunk embeddings stored in ChromaDB (or pgvector)                              | Vector search, embedding search          |
| <b>Keyword Search</b>  | BM25-based lexical search over tokenized and stemmed chunk text                                              | BM25 search, text search, lexical search |
| <b>Hybrid Search</b>   | RRF fusion of semantic and keyword search results into a single ranked list                                  | Combined search, dual search             |

Table 2 – continued

| Term                 | Definition                                                                                          | Aliases to avoid       |
|----------------------|-----------------------------------------------------------------------------------------------------|------------------------|
| <b>RRF</b>           | Reciprocal Rank Fusion —the algorithm that merges semantic and keyword rankings into unified scores | Fusion, rank merge     |
| <b>Stem Tags</b>     | The first 20 stemmed tokens of a chunk, stored as metadata to support downstream filtering          | Tags, keywords, tokens |
| <b>Search Result</b> | A single retrieved chunk with its score, source, chunk ID, and provenance metadata                  | Hit, result, match     |

## Security & Governance

| Term                                | Definition                                                                                                                                     | Aliases to avoid                                                                   |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Prompt Injection</b>             | A malicious query attempting to override system instructions or exfiltrate the soul                                                            | Injection attack, jailbreak                                                        |
| <b>Injection Filter</b>             | The regex-based input scanner that rejects queries matching banned patterns                                                                    | Prompt filter, sanitizer (the sanitizer is a separate concept)                     |
| <b>Soul</b>                         | The persistent identity/personality statement (from <code>soul.md</code> ) prepended to every LLM system prompt                                | Personality, identity, system prompt (the soul IS the system prompt additive)      |
| <b>Soul Evolution</b>               | The governed process of updating the soul, requiring human reason and passing injection scan                                                   | Soul update, personality change, soul mutation (mutation is the unauthorized form) |
| <b>Soul Drift</b>                   | When the on-disk <code>soul.md</code> differs from the last recorded version in the soul database                                              | Hash mismatch, file drift                                                          |
| <b>Soul Mutation</b>                | An unauthorized attempt to modify the soul via a normal query (blocked by rails)                                                               | Soul override, identity hijack                                                     |
| <b>Triple-Gated External Access</b> | The three conditions required for online escalation: hybrid mode + provider enabled + user confirmed                                           | Triple gate, external gate                                                         |
| <b>Out-of-Band</b>                  | A subsystem that runs strictly outside the request hot path (sync, agentic, <code>fsconnect</code> ) and never imports <code>gate/graph</code> | OOB, offline process, background task                                              |

Table 3 – continued

| Term                    | Definition                                                                                                                                                                  | Aliases to avoid              |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <b>Module Isolation</b> | The architectural rule that sync/ agentic/guardrails/fsconnect/ sqlconnect never import <code>gate.py</code> , <code>graph.py</code> , or <code>mcp_hybrid_server.py</code> | Separation, decoupling        |
| <b>Fail-Closed</b>      | A security design where any error or misconfiguration results in denial/blocking rather than permission                                                                     | Safe default, deny-by-default |
| <b>Pathsafe</b>         | The directory-fd-based security core that provably contains filesystem access within allowed roots                                                                          | Path validation, containment  |

## LLM & Generation

| Term                               | Definition                                                                                              | Aliases to avoid                       |
|------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Local LLM</b>                   | The LM Studio instance serving OpenAI-compatible chat completions on localhost                          | LM Studio, local model, on-premise LLM |
| <b>Grok</b>                        | The xAI cloud LLM used as the first online fallback provider                                            | xAI, external model (ambiguous)        |
| <b>Claude</b>                      | The Anthropic cloud LLM used as the second online fallback provider                                     | Anthropic, external model (ambiguous)  |
| <b>Retrieval-Before-Generation</b> | The core invariant: every answer must be grounded in retrieved context before any LLM generation occurs | RAG-first, context-first               |
| <b>Grounding Score</b>             | Token-overlap ratio between answer and retrieved context; a model-free faithfulness proxy               | Faithfulness score, overlap score      |
| <b>Hallucination</b>               | An answer containing claims unsupported by retrieved context                                            | Fabrication, ungrounded answer         |

## Data & Storage

| Term                  | Definition                                                                                | Aliases to avoid              |
|-----------------------|-------------------------------------------------------------------------------------------|-------------------------------|
| <b>Semantic Index</b> | The vector embedding store (ChromaDB or pgvector) containing chunk embeddings             | Vector index, embedding store |
| <b>BM25 Index</b>     | The JSON-serialized BM25 data structure containing tokenized corpus, chunks, and metadata | Keyword index, text index     |

Table 5 – continued

| Term                   | Definition                                                                                           | Aliases to avoid                   |
|------------------------|------------------------------------------------------------------------------------------------------|------------------------------------|
| <b>Soul Versions</b>   | The append-only audit table recording every soul change with SHA-256, content, reason, and timestamp | Soul history, personality log      |
| <b>Audit Log</b>       | The JSONL event stream recording every query, escalation, error, and governance action               | Event log, access log              |
| <b>Query Hash</b>      | The SHA-256 digest of query text stored in the audit log instead of the raw query                    | Hash, query fingerprint            |
| <b>Skills Registry</b> | The JSON-backed catalog of governed skills (name + description + body) with version history          | Skill catalog, capability registry |

## System & Architecture

| Term                  | Definition                                                                                                                                    | Aliases to avoid                    |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>Gateway</b>        | The FastAPI HTTP server ( <code>gate.py</code> ) handling <code>/query</code> , <code>/health</code> , <code>/soul</code> , and ops endpoints | API server, HTTP layer, FastAPI app |
| <b>Graph</b>          | The LangGraph state machine ( <code>graph.py</code> ) controlling query routing through retrieval, scoring, generation, and audit nodes       | State machine, router, pipeline     |
| <b>Graph Topology</b> | The edge structure of the LangGraph that deterministically enforces routing policy                                                            | Graph edges, routing structure      |
| <b>Health Probe</b>   | A runtime check of external dependency availability (LM Studio, Grok API, Claude API)                                                         | Health check, status check          |
| <b>Rate Limiter</b>   | The sliding-window request throttle per client IP, with optional SQLite/Postgres persistence                                                  | Throttle, request limiter           |
| <b>Ops Runner</b>     | The subprocess boundary that lets the gateway invoke out-of-band CLIs without importing them                                                  | Ops shim, subprocess wrapper        |
| <b>Config</b>         | The single <code>config.yaml</code> file that is the source of truth for all tunables                                                         | Configuration, settings             |

## Connectors & Integrations

| Term                    | Definition                                                                                               | Aliases to avoid                       |
|-------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------|
| <b>Sync</b>             | The rclone-based Dropbox corpus synchronization subsystem (pull or bisync)                               | Dropbox sync, corpus sync, rclone sync |
| <b>Agentic</b>          | The GitHub context and skills registry subsystem for human-governed automation                           | GitHub agent, agent layer              |
| <b>FSConnect</b>        | The filesystem connector providing scoped read/write access to local directories                         | File connector, filesystem layer       |
| <b>SQLConnect</b>       | The read-only SQL connector for database introspection                                                   | Database connector, SQL layer          |
| <b>Scoped Roots</b>     | The allowed directory list that confines FSConnect operations via held directory file descriptors        | Root list, allowed directories         |
| <b>Write Gate</b>       | The four-condition gate (mode + writes_enabled + reason + confirm) that controls all mutating operations | Mutation gate, write barrier           |
| <b>Governance Score</b> | A 0-100 rating of how well-governed a skill or soul proposal is (low injection risk, good structure)     | Quality score, trust score             |

## Relationships

- A **Query** passes through the **Score Gate**; if the top **Search Result** score is  $\geq \text{min\_score}$ , it becomes a **Local Answer**; otherwise it requires **User Confirmation** for **Online Escalation**
- A **Corpus** is split into **Chunks**, each stored in both the **Semantic Index** (embeddings) and the **BM25 Index** (tokenized text)
- **Hybrid Search** produces **Search Results** by fusing **Semantic Search** and **Keyword Search** via **RRF**
- The **Soul** is prepended to every **Local LLM** system prompt; **Soul Evolution** requires a human reason and passes the **Injection Filter**
- **Soul Drift** is detected when the on-disk soul.md hash differs from the latest **Soul Versions** row
- **Sync**, **Agentic**, **FSConnect**, **SQLConnect**, and **Guardrails** are all **Out-of-Band** subsystems that preserve **Module Isolation**
- The **Ops Runner** is the only path by which the **Gateway** can invoke **Out-of-Band** subsystems
- A **Skills Registry** entry has a **Governance Score** and must pass the same **Injection Filter** as the **Soul**
- **FSConnect** reads and writes are confined to **Scoped Roots** via the **Pathsafe** security core
- **Audit Log** records every **Query** as a **Query Hash**, never raw text (when include\_query\_hash is enabled)

## Example dialogue

**Dev:** “When a **Query** hits the **Score Gate** with a low top **Search Result** score, what happens?”  
**Domain expert:** “The **Graph** routes it to the user confirmation node. If the user confirms, it checks which **Online Escalation** provider is selected — **Grok** or **Claude** — and forwards the query there. If they decline, it falls back to **Offline Best-Effort**.” **Dev:** “And the **Audit Log** captures which path was taken?” **Domain expert:** “Exactly — every routing decision, the **Retrieval Mode**, the **Query Hash**, and which model produced the answer. The raw query text never lands in the log.”  
**Dev:** “What about **Soul Evolution** — how does that differ from a normal **Soul Drift**?” **Domain expert:** “**Soul Drift** is an unplanned mismatch detected on startup or reload — the system adopts the file but audits it. **Soul Evolution** is the governed path: a human provides a reason, the proposal passes the **Injection Filter**, and only then does `apply_evolution` atomically write the new **Soul** and append a row to **Soul Versions**.” **Dev:** “Can the **Agentic** layer write to GitHub directly?”  
**Domain expert:** “No — v0.1 has **EXECUTION\_ENABLED** hardcoded to `False`. Even with all four **Write Gate** conditions satisfied, `plan_write` only returns a dry-run argv plan. Real writes require a separate, reviewed code change.”

## Flagged ambiguities

- “**external fallback**” was used to mean both **Grok** and **Claude** collectively, but also specifically the Grok path. Prefer “**Online Escalation**” for the generic concept and the provider name for the specific one.
- “**sanitizer**” (verb form) and “**injection filter**” (noun form) are distinct: `sanitize_chunk` replaces banned patterns with `[FILTERED]` at index time, while `check_input` raises `PromptInjectionError` at query time. Both use the same config-driven pattern set, but their failure modes differ.
- “**mode**” has three meanings: `app.mode` (“offline” vs “hybrid”), `agentic.mode` (“read” vs “write”), and `retrieval_mode` (“semantic” / “keyword” / “hybrid”). Always qualify with the subsystem prefix.
- “**config**” refers specifically to `config.yaml` as the single source of truth, not to any individual subsystem’s parsed dataclass (`AgenticConfig`, `RcloneConfig`, `FsConnectConfig`). Prefer “config block” for the YAML subsection and “config object” for the parsed form.
- “**guardrails**” refers to the NeMo Guardrails integration skeleton (future path), while “**injection filter**” and “**rails**” in `guardrails/rails.py` refer to the offline Python check functions that work without NeMo. The two share the same logic but different activation paths.
- “**score**” is overloaded: `SearchResult.score` may be raw BM25, raw cosine similarity, or RRF-fused depending on `retrieval_mode`. Always check the `retrieval_mode` field before interpreting a score value.